

Applications of IC

Applications of the IC characterization

- Revenue equivalence.
- Obtaining equilibria in specific auctions.
- Optimal reserve prices and entry fees.
- Myerson (1981), ($N = 1$, I bidders)
 - Alternative expression for revenue.
 - Optimal mechanism.
 - Optimal reserve price in classic auctions.

Revenue equivalence

Theorem 1 (Revenue equivalence) IPV with I bidders. $X_i = [\underline{x}_i, \bar{x}_i]$. Let (p, t) and (p', t') be two DRMs and let $(Q, T), (Q', T')$ be the corresponding expected probability of trade and expected transfer functions. If

$$(\forall i) Q_i = Q'_i, \text{ then } T_i = T'_i + [U'_i(\underline{x}_i) - U_i(\underline{x}_i)].$$

- This formulation of *revenue equivalence* and its proof in the next slide are due to Williams (1999).
- The same statement and proof hold in the extension of the model to N goods.

Proof. By definition,

$$T_i(x_i) = \frac{dU_i(x_i)}{dx_i} \cdot x_i - U_i(x_i)$$
$$T'_i(x_i) = \frac{dU'_i(x_i)}{dx_i} \cdot x_i - U'_i(x_i).$$

- By BIC, $\frac{dU_i}{dx_i} = Q_i = Q'_i = \frac{dU'_i}{dx_i}$ a.e.
- Then, $T_i(x_i) = T'_i(x_i) + [U'_i(x_i) - U_i(x_i)]$ a.e.
- If two convex functions have the same gradient a.e., then both functions are equal up to a constant Rockafellar (1970), Chapter 24. Hence,

$$(\forall x_i) [U'_i(x_i) - U_i(x_i)] \text{ is a constant.}$$

QED

Revenue equivalence

Corollary 1 The four classic auctions yield the same expected revenue to the seller.

Proof. At the respective equilibria of the four classic auctions, every bidder i faces $Q_i(x_i) = F(x_i)^{I-1}$ and $U_i(\underline{x}_i) = 0$. QED

First-price sealed-bid auction

Theorem 2 (FPA) Symmetric IPV, $N = 1$ good, I bidders. The unique, symmetric, monotone Bayesian-Nash equilibrium of the FPA is

$$(\forall i), \quad b_i(x_i) = x_i - \int_0^{x_i} \left(\frac{F(y)^{I-1}}{F(x_i)^{I-1}} \right) dy.$$

Proof. Sketch.

1. If the equilibrium strategy is monotone nondecreasing then

$$Q_i(x_i) = F(x_i)^{I-1}.$$

2. Under the assumption of monotonicity, we have the system

$$U_i(x_i) = F(x_i)^{I-1}x_i - T_i(x_i)$$

$$U_i(x_i) = U_i(0) + \int_0^{x_i} F(y)^{I-1} \, dy$$

$$T_i(x_i) = F^{I-1}(x_i)b(x_i)$$

3. Solve for $b(x_i)$ and check it is monotone.

Reserve price and entry fees

- Before bidding agents pay an entry fee c . Only agents that choose to submit bids pay c .
- Generally $c > 0$. Sometimes $c < 0$ to compensate bidders for cost of bid preparation and to increase participation.
- A reserve price r is a minimum price chosen by the seller. It may be announced or kept secret.
If no bid exceeds r the seller keeps the object.
- In SPA, if only one bid exceeds r , the bidder gets the object and pays r .

SPA: Reserve price and entry fee

- SIPVM, $X_i = [0, 1]$. SPA with entry fee c and reserve price r .
- Buyers' expected profits from bidding are negative if $c > 1 - r$.
- Suppose $0 \leq c \leq 1 - r \leq 1$. Find a non-trivial equilibrium of the SPA.
- A participant will only bid if her valuation exceeds a *marginal value* x'_i .
The marginal value x'_i can be determined intuitively by considering the incentives of a bidder with valuation x'_i . Then, use x'_i to guess an equilibrium strategy.

References

- Myerson, Roger. 1981. "Optimal Auction Design." *Mathematics of Operations Research* 6: 58–73.
- Rockafellar, R. Tyrrell. 1970. *Convex Analysis*. Number 28 in Princeton Mathematical Series. Princeton University Press.
- Williams, Steven. 1999. "A Characterization of Efficient, Bayesian Incentive Compatible Mechanisms." *Economic Theory* 14: 155–80.